ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PRACTICE EXAM 6 (PAPER 2)

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET second paper (practical, 80 marks): breadboard prototypes of a dark-activated switch, a two-hand safety circuit with an AND gate IC, a water-level alarm, and a DC motor speed controller with its circuit diagram.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.
2. This is a practical paper. Use Fritzing for breadboard prototypes and draw.io (or similar) for circuit diagrams. If you do not have a computer, draw the breadboard layouts and circuit diagrams neatly by hand.
3. Put your name or examination number at the top of every printout or page.
4. Show the positive and negative supply rails clearly, place every component correctly and make all the connections.
5. Save your work regularly.

QUESTION 1

The circuit below switches on an LED when it gets dark, for example to light up a robot's work area at night. Design and print its breadboard prototype.

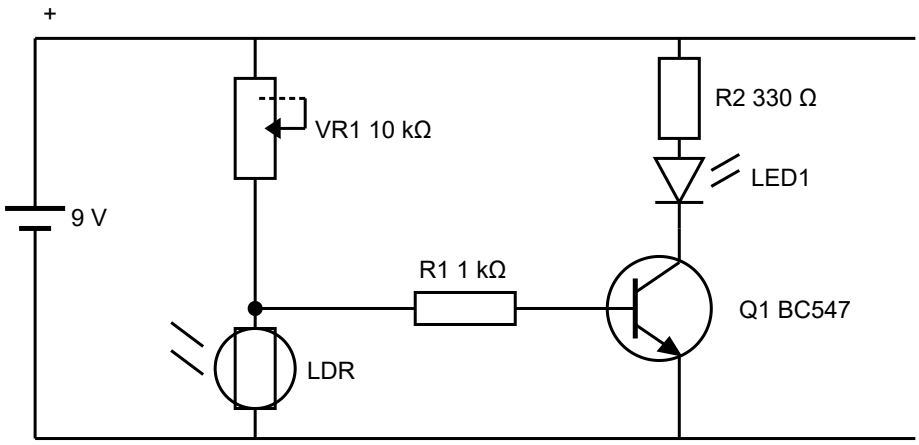


FIGURE 1: Dark-activated switch

Components: 1 × 9 V battery; 1 × LDR; 1 × 10 kΩ potentiometer (VR1); 1 × 1 kΩ resistor (R1); 1 × 330 Ω resistor (R2); 1 × BC547 NPN transistor; 1 × LED; breadboard and jumper wires.

Design and print (or draw) the breadboard prototype of the circuit in FIGURE 1. (20)

[20]

QUESTION 2

A workshop press may only move when the operator presses TWO buttons at the same time, one with each hand, so that both hands are away from the danger area. The circuit below uses one gate of a 74HC08 AND gate IC. The LED shows that the press may move.

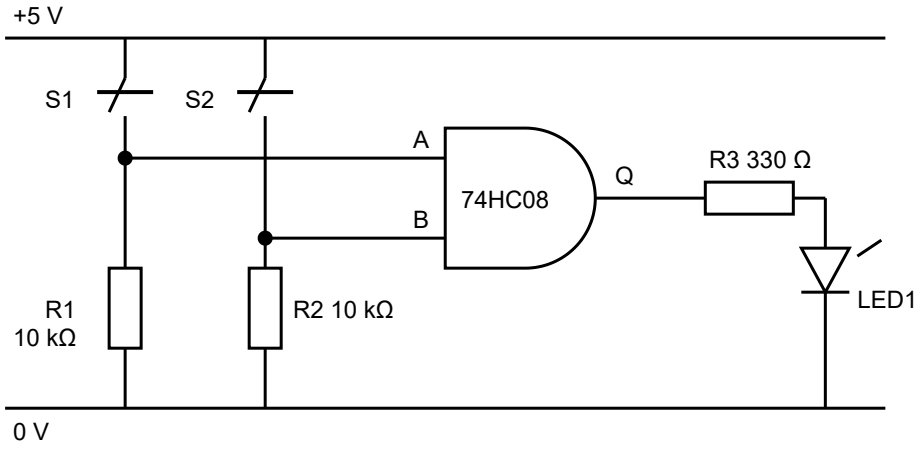


FIGURE 2: Two-hand safety control (IC pin 14 to +5 V and pin 7 to 0 V are not shown)

Components: 1 × 5 V supply; 1 × 74HC08 quad AND gate IC; 2 × push buttons (S1, S2); 2 × 10 kΩ resistors (R1, R2); 1 × 330 Ω resistor (R3); 1 × green LED; breadboard and jumper wires.

Design and print (or draw) the breadboard prototype of the circuit in FIGURE 2. Use gate 1 of the IC (inputs on pins 1 and 2, output on pin 3) and remember the IC's supply pins. (20)

[20]

QUESTION 3

Lerato's family stores rain water in a tank, and it sometimes overflows. She wants a circuit that sounds a buzzer and lights a red LED when the water reaches the top of the tank. Two bare wire probes are fixed near the top of the tank: when the water touches both probes, a small current flows through the water into the base of a transistor, which switches on the buzzer and the LED.

Components: 1 × 9 V battery; 2 × wire probes (shown as two connector pins or wires); 1 × BC547 NPN transistor; 1 × 10 kΩ resistor (base); 1 × 9 V buzzer; 1 × red LED; 1 × 470 Ω resistor; breadboard and jumper wires.

Use Fritzing (or draw by hand) to design the breadboard prototype of Lerato's water-level alarm. The buzzer, and the LED with its resistor, must be in parallel in the collector circuit of the transistor. Add your name or examination number and print. (20)

[20]

QUESTION 4

Sipho must build a speed controller for a 12 V DC motor that drives a robot's conveyor belt. A 10 kΩ potentiometer sets the base current of a TIP120 Darlington transistor, which controls the current through the motor. A 1N4007 diode across the motor protects the transistor from voltage spikes when the motor is switched off.

Components: 1 × 12 V DC supply; 1 × 12 V DC motor; 1 × TIP120 Darlington transistor; 1 × 10 kΩ potentiometer; 1 × 1 kΩ resistor (base); 1 × 1N4007 diode; 1 × red LED with a 1 kΩ resistor (power-on indicator); breadboard and jumper wires.

4.1 Draw the circuit diagram of the motor speed controller (in draw.io or neatly by hand), using the correct symbols. (5)

4.2 Design the breadboard prototype of the motor speed controller in Fritzing (or draw it by hand). Add your name or examination number and print. (15)

[20]

TOTAL: 80

END OF QUESTION PAPER